



Lowest common multiple

Task A: Common multiples using prime factorisation

1) Write the first eight multiples of the following:

a) 3×5 a) $3 \times 5, 2 \times 3 \times 5, 3^2 \times 5, 2^2 \times 3 \times 5, 3 \times 5^2,$
 15 $2 \times 3^2 \times 5, 3 \times 5 \times 7, 2^3 \times 3 \times 5$

b) $2^2 \times 3$ b) $2^2 \times 3, 2^3 \times 3, 2^2 \times 3^2, 2^4 \times 3, 2^2 \times 3 \times 5, 2^3 \times 3^2,$
 12 $2^2 \times 3 \times 7, 2^5 \times 3$

c) $2 \times 3^2 \times 5$ c) $2 \times 3^2 \times 5, 2^2 \times 3^2 \times 5, 2 \times 3^3 \times 5, 2^3 \times 3^2 \times 5,$
 $2 \times 3^2 \times 5^2, 2^2 \times 3^3 \times 5, 2 \times 3^2 \times 5 \times 7, 2^4 \times 3^2 \times 5$

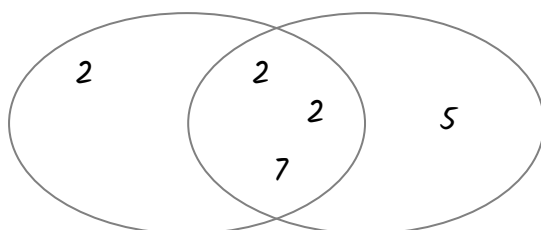
d) d) the common multiple is $2^2 \times 3 \times 5 = 60$

Task B: Lowest common multiple using prime factorisation

1) Find the **LCM** of the following pairs of numbers:

a) $56 = 2 \times 2 \times 2 \times 7$

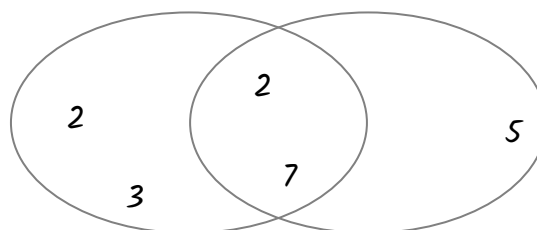
$140 = 2 \times 2 \times 5 \times 7$



The **LCM** of 56 and 140 is 280

b) $84 = 2 \times 2 \times 3 \times 7$

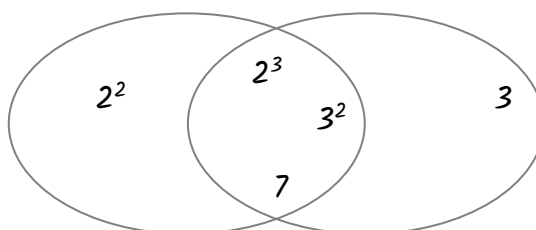
$70 = 2 \times 5 \times 7$



The **LCM** of 84 and 70 is 420

c) $2016 = 2^5 \times 3^2 \times 7$

$1512 = 2^3 \times 3^3 \times 7$

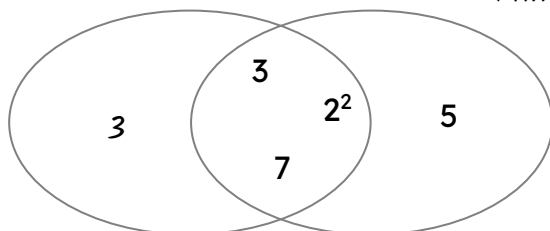


The **LCM** of 2016 and 1512 is 6048

2) The **LCM** of the 420 and another number is represented in the Venn diagram is 1260.

- What is the missing prime factor?
- What was the other number?

Prime factors of 420



- $2^2 \times 3 \times 5 \times 7 \times \boxed{3} = 1260$
- The other number is $2^2 \times 3^2 \times 7 = 252$

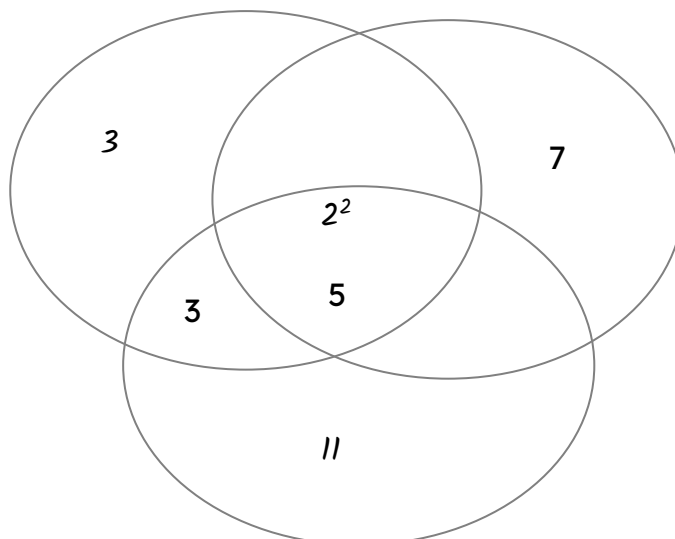
Task C: Lowest common multiple of more than two numbers

Fill in the blanks

$$180 = 2^2 \times 3^2 \times 5$$

$$140 = 2^2 \times 5 \times 7$$

$$660 = 2^2 \times 3 \times 5 \times 11$$



The **LCM** of 180, 140 and 660 is 13 860

